EE671: VLSI Design

Course Project 1

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1 Introduction

This is the course project of the course EE671: VLSI Design. This project focuses on the design, layout, extraction, and verification of CMOS standard cells using the SKY130A process technology. The cells assigned to us were:

1. buf: Buffer
2. nor2: 2-input NOR gate
3. dfxtp: Delay flop, single output Q, posedge clk

For each cell provided, the circuit was designed by hand using CMOS transistors . We then made NGSpice netlists for them and simulated them to get required rise and fall times which matched the inverter specifications from the assignments before. Following this, design layouts were created on magic using the Sky130A process technology, keeping in mind the Design Rule Check (DRC). LVS was checked and LEF was extracted, power analysis was completed. Lastly, .lib files was generated using charlib as part of the bonus.

2 Aim

The aim of this project is to design and physically implement CMOS standard cells using the SKY130A technology and verify their correctness through simulation and layout extraction. The specific objectives of the project are:

1. To design a 2-input CMOS NOR gate and an inverter at the transistor level.
2. To determine suitable transistor dimensions for the required circuit performance.
3. To simulate the designed circuits using `ngspice` and obtain parameters such as rise time, fall time, and propagation delay.
4. To create the corresponding physical layouts in Magic VLSI using the SKY130A process design kit.
5. To ensure that the layouts satisfy the relevant design rules through Design Rule Check (DRC).
6. To extract the transistor-level netlists from the layouts.
7. To compare the extracted SPICE netlists with the original schematic/netlists and verify the connectivity and device implementation of the physical designs.

3 Combinational-1: Buffer

3.1 Schematic design

Table 1: Buffer Rise and Fall Times from Schematic Simulation

Parameter	Value
Rise time t_r	18.34 ps
Fall time t_f	18.53 ps

```
*Buffer (BUF_1X) - two-stage non-inverting buffer, sized to match INVX1 rise/fall

.lib /foss/pdks/sky130A/libs.tech/ngspice/sky130.lib.spice tt


.subckt inv1 a vdd vss z
xm01 z a vdd vdd sky130_fd_pr__pfet_01v8
+ l=0.15 w=1.25
+ as=0.375 ad=0.375
+ ps=3.1 pd=3.1

xm02 z a vss vss sky130_fd_pr__nfet_01v8
+ l=0.15 w=0.42
+ as=0.126 ad=0.126
+ ps=1.44 pd=1.44
.ends inv1

.subckt buf a vdd vss z

* Stage 1
xm11 mid a vdd vdd sky130_fd_pr__pfet_01v8
+ l=0.15 w=1.42
+ as=0.426 ad=0.426
+ ps=3.44 pd=3.44

xm12 mid a vss vss sky130_fd_pr__nfet_01v8
+ l=0.15 w=0.47
+ as=0.141 ad=0.141
+ ps=1.54 pd=1.54

* Stage 2 (output stage, = INVX1 size)
xm21 z mid vdd vdd sky130_fd_pr__pfet_01v8
+ l=0.15 w=1.42
+ as=0.426 ad=0.426
+ ps=3.44 pd=3.44

xm22 z mid vss vss sky130_fd_pr__nfet_01v8
+ l=0.15 w=0.47
+ as=0.141 ad=0.141
+ ps=1.54 pd=1.54

.ends buf

*-------------------------------------------------------
* Testbench: BUF loaded with INVX1 (same fanout condition
```

```
* used to characterize INVX1 in Assignment 1)
*-------------------------------------------------------
Vdd vdd 0 DC 1.8
Vin in 0 PULSE(0 1.8 0 20p 20p 1n 2n)

XBUF1 in vdd 0 out buf
XLOAD out vdd 0 output inv1

.tran 1p 10n

.control
run
plot V(in) V(out)

meas tran trise TRIG v(out) VAL=0.36 RISE=1
+ TARG v(out) VAL=1.44 RISE=1

meas tran tfall TRIG v(out) VAL=1.44 FALL=1
+ TARG v(out) VAL=0.36 FALL=1

meas tran tphl TRIG v(in) VAL=0.9 RISE=1
+ TARG v(out) VAL=0.9 RISE=1

meas tran tplh TRIG v(in) VAL=0.9 FALL=1
+ TARG v(out) VAL=0.9 FALL=1
.endc
.end
```

3.2 Circuit diagram

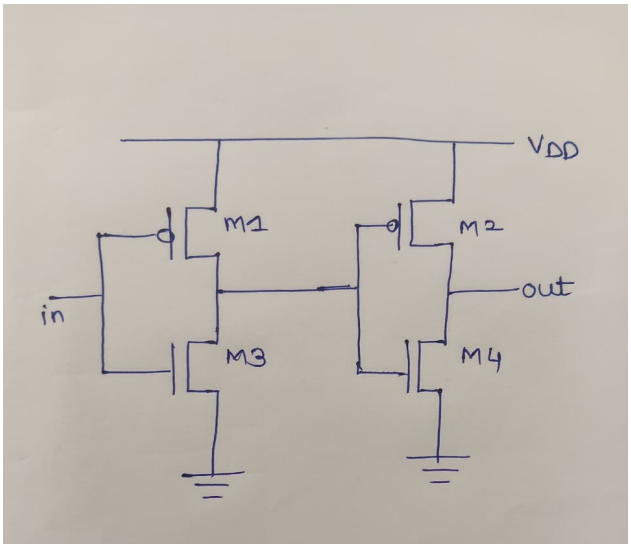


Figure 1: Buffer circuit

Table 2: MOSFET Widths for the Buffer Circuit

MOSFET	Width (μm)
M1 and M2 (PMOS)	1.42 um
M3 and M4 (NMOS)	0.47 um

3.3 Magic Layout (DRC clean)

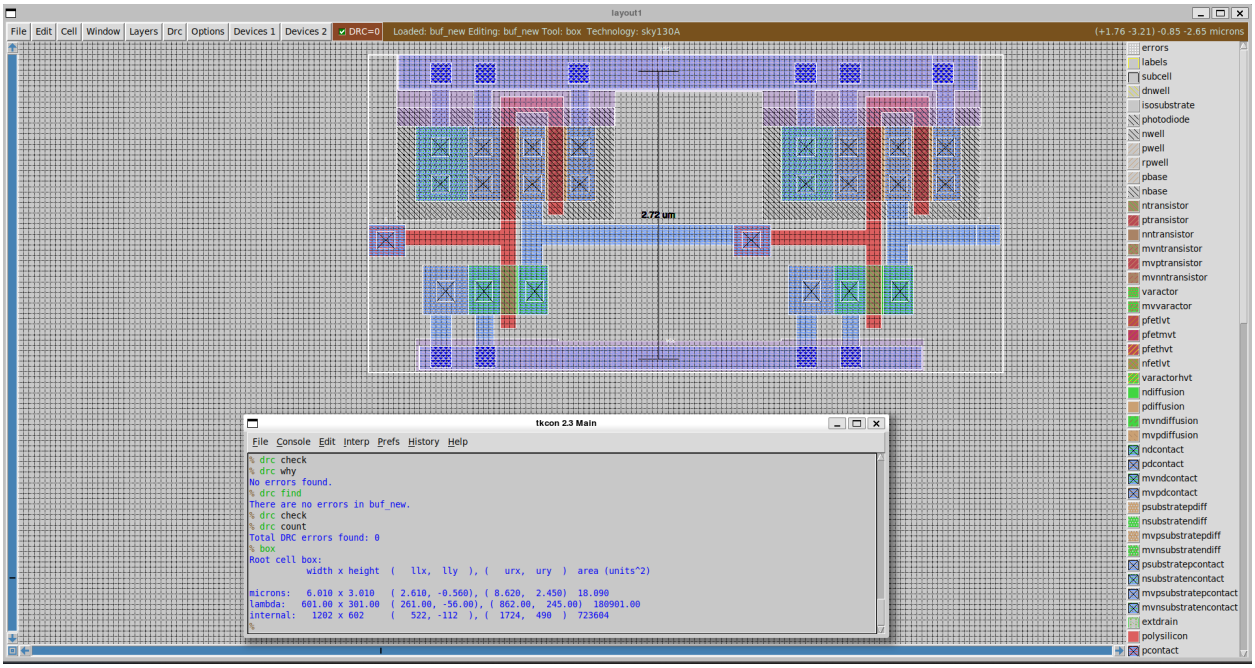


Figure 2: Magic Layout of Buffer Cell with width and height

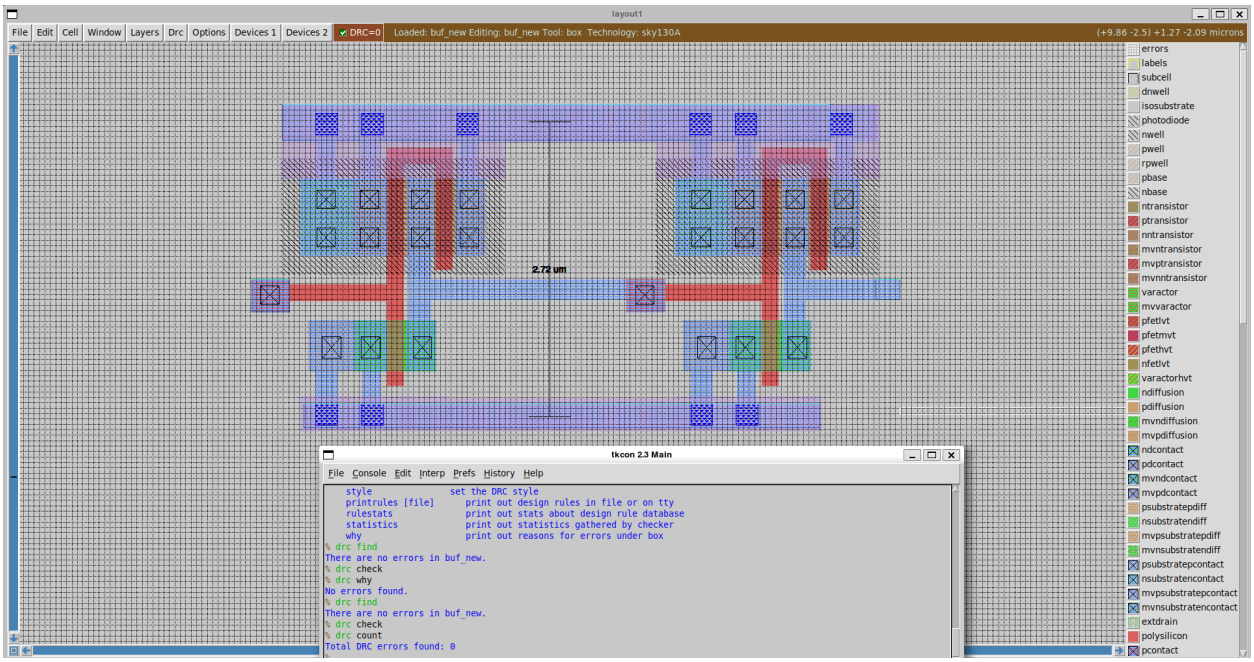


Figure 3: Magic Layout of Buffer Cell with DRC Check

3.4 LVS check

```
/foss/designs > netgen -batch lvs "buf_new_layout.spice buf_new" "buf_new.spice buf" /foss/pdks/sky130A/libs.tech/netgen/sky130A_setup.tcl
Netgen 1.5.323 compiled on Mon Jul 27 09:16:59 AM CEST 2026
Warning: netgen command 'format' use fully-qualified name '::netgen::format'
Warning: netgen command 'global' use fully-qualified name '::netgen::global'
Reading netlist file buf_new_layout.spice for buf_new_layout.spice buf_new
Call to undefined subcircuit sky130_fd_pr__nfet_01v8
Creating placeholder cell definition.
Call to undefined subcircuit sky130_fd_pr__pfet_01v8
Creating placeholder cell definition.
Reading netlist file buf_new.spice for buf_new.spice buf
Ignoring line starting with token: .lib
line number 3 = '.lib /foss/pdks/sky130A/libs.tech/ngspice/sky130.lib.spice tt'
Call to undefined subcircuit sky130_fd_pr__pfet_01v8
Creating placeholder cell definition.
Call to undefined subcircuit sky130_fd_pr__nfet_01v8
Creating placeholder cell definition.
Ignoring line starting with token: .tran
line number 54 = '.tran 1p 10n'
File buf_new.spice read with 2 warnings.

Reading setup file /foss/pdks/sky130A/libs.tech/netgen/sky130A_setup.tcl

Model sky130_fd_pr__nfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__nfet_01v8
No property sa found for device sky130_fd_pr__nfet_01v8
No property sb found for device sky130_fd_pr__nfet_01v8
No property sd found for device sky130_fd_pr__nfet_01v8
No property nf found for device sky130_fd_pr__nfet_01v8
No property nrd found for device sky130_fd_pr__nfet_01v8
No property nrs found for device sky130_fd_pr__nfet_01v8
No property area found for device sky130_fd_pr__nfet_01v8
No property perim found for device sky130_fd_pr__nfet_01v8
No property topography found for device sky130_fd_pr__nfet_01v8
Model sky130_fd_pr__nfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__nfet_01v8
No property sa found for device sky130_fd_pr__nfet_01v8
No property sb found for device sky130_fd_pr__nfet_01v8
No property sd found for device sky130_fd_pr__nfet_01v8
No property nf found for device sky130_fd_pr__nfet_01v8
No property nrd found for device sky130_fd_pr__nfet_01v8
No property nrs found for device sky130_fd_pr__nfet_01v8
No property area found for device sky130_fd_pr__nfet_01v8
No property perim found for device sky130_fd_pr__nfet_01v8
No property topography found for device sky130_fd_pr__nfet_01v8
Model sky130_fd_pr__pfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__pfet_01v8
No property sa found for device sky130_fd_pr__pfet_01v8
No property sb found for device sky130_fd_pr__pfet_01v8
No property sd found for device sky130_fd_pr__pfet_01v8
No property nf found for device sky130_fd_pr__pfet_01v8
No property nrd found for device sky130_fd_pr__pfet_01v8
No property nrs found for device sky130_fd_pr__pfet_01v8
```

```
No property area found for device sky130_fd_pr__pfet_01v8
No property perim found for device sky130_fd_pr__pfet_01v8
No property topography found for device sky130_fd_pr__pfet_01v8
Model sky130_fd_pr__pfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__pfet_01v8
No property sa found for device sky130_fd_pr__pfet_01v8
No property sb found for device sky130_fd_pr__pfet_01v8
No property sd found for device sky130_fd_pr__pfet_01v8
No property nf found for device sky130_fd_pr__pfet_01v8
No property nrd found for device sky130_fd_pr__pfet_01v8
No property nrs found for device sky130_fd_pr__pfet_01v8
No property area found for device sky130_fd_pr__pfet_01v8
No property perim found for device sky130_fd_pr__pfet_01v8
No property topography found for device sky130_fd_pr__pfet_01v8
Comparison output logged to file comp.out
Logging to file "comp.out" enabled
Circuit sky130_fd_pr__nfet_01v8 contains no devices.
Circuit sky130_fd_pr__pfet_01v8 contains no devices.

Contents of circuit 1:  Circuit: 'buf_new'
Circuit buf_new contains 6 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:   4
  Class: sky130_fd_pr__nfet_01v8 instances:   2
Circuit contains 5 nets.
Contents of circuit 2:  Circuit: 'buf'
Circuit buf contains 4 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:   2
  Class: sky130_fd_pr__nfet_01v8 instances:   2
Circuit contains 5 nets.

Circuit was modified by parallel/series device merging.
New circuit summary:

Contents of circuit 1:  Circuit: 'buf_new'
Circuit buf_new contains 4 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:   2
  Class: sky130_fd_pr__nfet_01v8 instances:   2
Circuit contains 5 nets.
Contents of circuit 2:  Circuit: 'buf'
Circuit buf contains 4 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:   2
  Class: sky130_fd_pr__nfet_01v8 instances:   2
Circuit contains 5 nets.

Circuit 1 contains 4 devices, Circuit 2 contains 4 devices.
Circuit 1 contains 5 nets,    Circuit 2 contains 5 nets.


Final result:
Circuits match uniquely.
.
Logging to file "comp.out" disabled
LVS Done.
```

3.5 PEX

```
* NGSPICE file created from buf_new.ext - technology: sky130A

.subckt buf_new vdd a z vss
X0 z a_802_n2# vss vss sky130_fd_pr__nfet_01v8 ad=0.141 pd=1.54 as=0.141 ps=1.54 w=0.47 l=0.15
**devattr s=5640,308 d=5640,308
X1 a_802_n2# a vss vss sky130_fd_pr__nfet_01v8 ad=0.141 pd=1.54 as=0.141 ps=1.54 w=0.47 l=0.15
**devattr s=5640,308 d=5640,308
X2 vdd a_802_n2# z vdd sky130_fd_pr__pfet_01v8 ad=0.213 pd=2.02 as=0.1065 ps=1.01 w=0.71 l=0.15
**devattr s=4260,202 d=8520,404
X3 z a_802_n2# vdd vdd sky130_fd_pr__pfet_01v8 ad=0.1065 pd=1.01 as=0.213 ps=2.02 w=0.71 l=0.15
**devattr s=8520,404 d=4260,202
X4 vdd a a_802_n2# vdd sky130_fd_pr__pfet_01v8 ad=0.213 pd=2.02 as=0.1065 ps=1.01 w=0.71 l=0.15
**devattr s=4260,202 d=8520,404
X5 a_802_n2# a vdd vdd sky130_fd_pr__pfet_01v8 ad=0.1065 pd=1.01 as=0.213 ps=2.02 w=0.71 l=0.15
**devattr s=8520,404 d=4260,202
C0 vdd a_802_n2# 0.3913f
C1 a_802_n2# z 0.05357f
C2 a a_802_n2# 0.05797f
C3 vdd z 0.1829f
C4 vdd a 0.18001f
C5 a z 0
C6 z vss 0.5575f
C7 vdd vss 1.60342f
.ends
```

3.6 Waveform

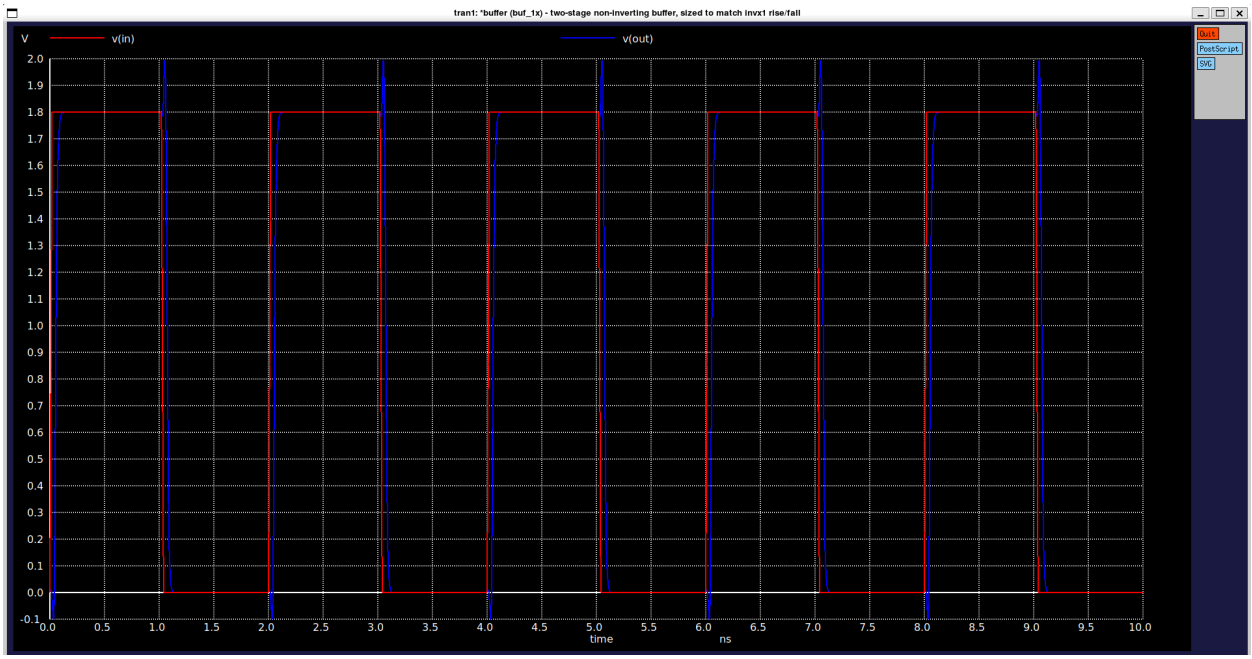


Figure 4: Buffer input and output waveforms showing rise and fall swings and propagation delays

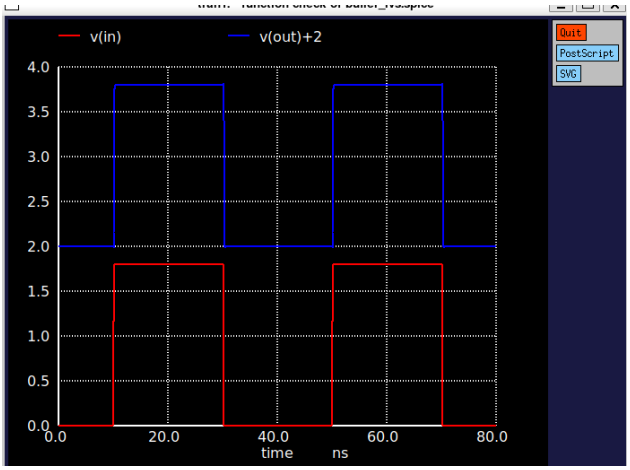
3.7 Testbench Simulation

```
/foss/designs > cat tb_buffer.spice
* Function check of buffer_lvs.spice
.lib $PDK_ROOT/sky130A/libs.tech/combined/sky130.lib.spice tt
.include buffer_schematic.spice

Vdd VDD 0 1.8
XDUT IN OUT VDD 0 buffer

Vin IN 0 pulse(0 1.8 10n 0.1n 0.1n 20n 40n)
Cl OUT 0 10f

.tran 0.1n 80n
.control
run
plot v(IN) v(OUT)+2
.endc
.end
```



3.8 Capacitance considerations

Table 3: Input Pin Capacitance for the Buffer

Input Pin	Rise Cap (pF)	Fall Cap (pF)	Average Cap (pF)
A	0.002688	0.002687	0.002688

3.9 Timing considerations

Table 4: Output rise time (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.01226	0.01254	0.02818
10 fF	0.05787	0.05791	0.06635
100 fF	0.51548	0.51548	0.51618

Table 5: Output fall time (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.01195	0.01231	0.02540
10 fF	0.06165	0.06170	0.06986
100 fF	0.54837	0.54837	0.54958

Table 6: Cell rise propagation delay (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.04237	0.06064	0.13228
10 fF	0.07821	0.09635	0.18166
100 fF	0.38920	0.40753	0.49453

Table 7: Cell fall propagation delay (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.04134	0.06173	0.13231
10 fF	0.08494	0.10541	0.18794
100 fF	0.45875	0.47969	0.56534

3.10 Power considerations

Table 8: Input combination vs. Static power

A	Static power (nW)
0	1.765294
1	1.765294

Table 9: Dynamic rise power (nW) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	752.04	665.95	940.71
10 fF	3889.08	3793.68	4013.60
100 fF	33066.36	32982.12	33208.20

Table 10: Dynamic fall power (nW) vs. Input Slew and Load Capacitance

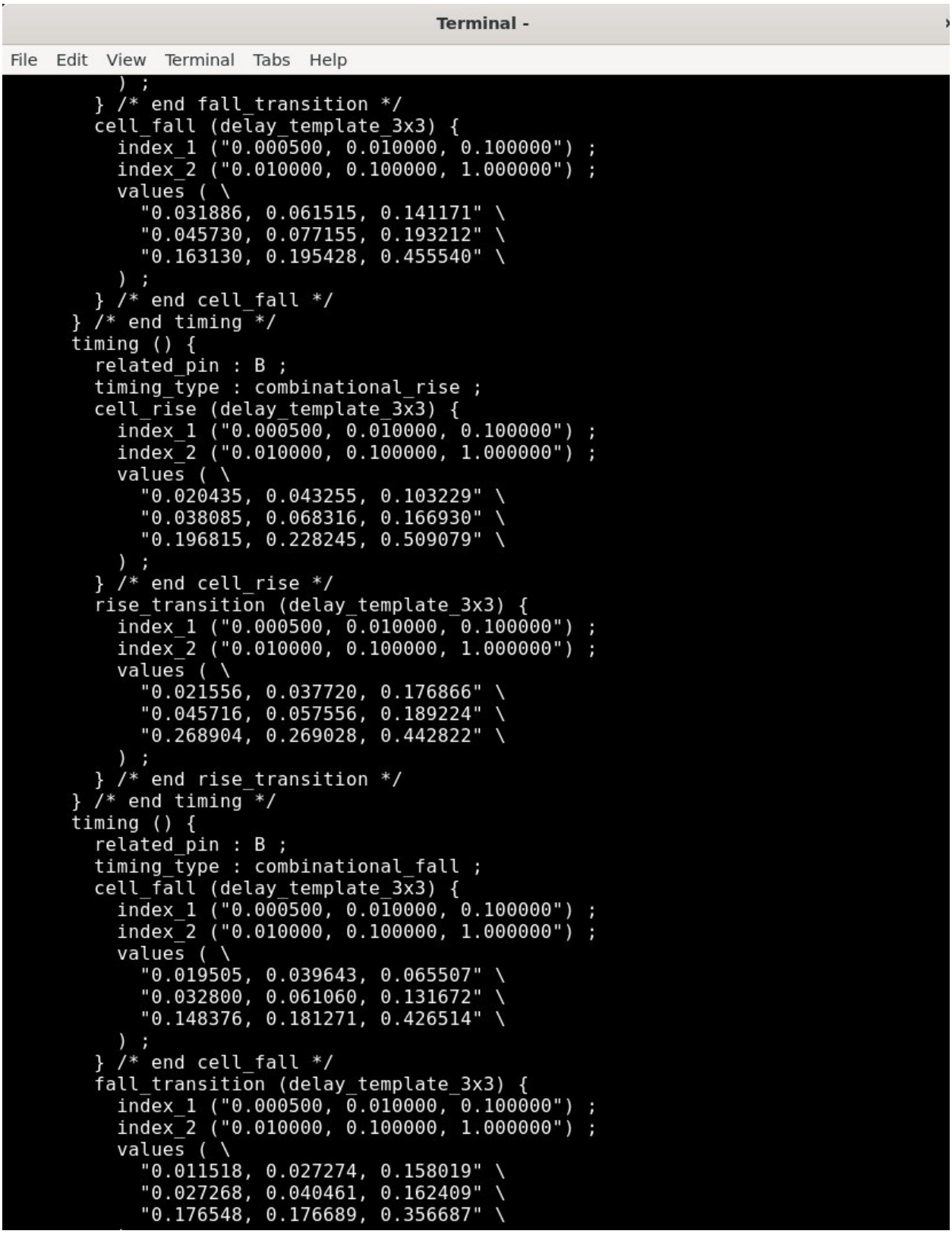
	10 ps	100 ps	1000 ps
0.5 fF	1552.75	1487.35	1864.91
10 fF	1630.66	1563.36	1894.95
100 fF	1648.95	1601.99	1965.94

3.11 CharLib

Terminal -

File Edit View Terminal Tabs Help

```
/foss/designs/charlib > grep -n "cell_rise\|cell_fall\|rise_transition\|fall_transi
tion" results/sky130_nor2.lib
49:        cell_rise (delay_template_3x3) {
57:        } /* end cell_rise */
58:        rise_transition (delay_template_3x3) {
66:        } /* end rise_transition */
71:        fall_transition (delay_template_3x3) {
79:        } /* end fall_transition */
80:        cell_fall (delay_template_3x3) {
88:        } /* end cell_fall */
93:        cell_rise (delay_template_3x3) {
101:        } /* end cell_rise */
102:        rise_transition (delay_template_3x3) {
110:        } /* end rise_transition */
115:        cell_fall (delay_template_3x3) {
123:        } /* end cell_fall */
124:        fall_transition (delay_template_3x3) {
132:        } /* end fall_transition */
/foss/designs/charlib > sed -n '45,135p' results/sky130_nor2.lib
      function : "Z = !(A|B)" ;
      timing () {
        related_pin : A ;
        timing_type : combinational_rise ;
        cell_rise (delay_template_3x3) {
          index_1 ("0.000500, 0.010000, 0.100000") ;
          index_2 ("0.010000, 0.100000, 1.000000") ;
          values ( \
            "0.033623, 0.063860, 0.124580" \
            "0.050971, 0.083634, 0.190486" \
            "0.209309, 0.242914, 0.496639" \
          ) ;
        } /* end cell_rise */
        rise_transition (delay_template_3x3) {
          index_1 ("0.000500, 0.010000, 0.100000") ;
          index_2 ("0.010000, 0.100000, 1.000000") ;
          values ( \
            "0.022535, 0.031199, 0.175642" \
            "0.045935, 0.050708, 0.171852" \
            "0.268904, 0.269038, 0.386913" \
          ) ;
        } /* end rise_transition */
      } /* end timing */
      timing () {
        related_pin : A ;
        timing_type : combinational_fall ;
        fall_transition (delay_template_3x3) {
          index_1 ("0.000500, 0.010000, 0.100000") ;
          index_2 ("0.010000, 0.100000, 1.000000") ;
          values ( \
            "0.022219, 0.033286, 0.157728" \
            "0.038663, 0.045453, 0.180177" \
            "0.188581, 0.188745, 0.363135" \
          ) ;
```


Terminal -
File Edit View Terminal Tabs Help
) ;
} /* end fall_transition */
cell_fall (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.031886, 0.061515, 0.141171" \
"0.045730, 0.077155, 0.193212" \
"0.163130, 0.195428, 0.455540" \
) ;
} /* end cell_fall */
} /* end timing */
timing () {
related_pin : B ;
timing_type : combinational_rise ;
cell_rise (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.020435, 0.043255, 0.103229" \
"0.038085, 0.068316, 0.166930" \
"0.196815, 0.228245, 0.509079" \
) ;
} /* end cell_rise */
rise_transition (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.021556, 0.037720, 0.176866" \
"0.045716, 0.057556, 0.189224" \
"0.268904, 0.269028, 0.442822" \
) ;
} /* end rise_transition */
} /* end timing */
timing () {
related_pin : B ;
timing_type : combinational_fall ;
cell_fall (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.019505, 0.039643, 0.065507" \
"0.032800, 0.061060, 0.131672" \
"0.148376, 0.181271, 0.426514" \
) ;
} /* end cell_fall */
fall_transition (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.011518, 0.027274, 0.158019" \
"0.027268, 0.040461, 0.162409" \
"0.176548, 0.176689, 0.356687" \

4 Combinational-2: 2-input NOR gate

4.1 Schematic Design

```
*NOR Gate Simulation
.lib /foss/pdks/sky130A/libs.tech/ngspice/sky130.lib.spice tt

.param l=0.15

* Inverter sizes
.param inv_wp=1.25
.param inv_wn=0.42

* NOR sizes
.param nor_wp=6
.param nor_wn=1.5

* Inverter ckt-
.subckt INVX1 A VDD VSS Z
xm01 Z A VDD VDD sky130_fd_pr__pfet_01v8 l={l} w={inv_wp} as={inv_wp*2*l} ad={i>
xm02 Z A VSS VSS sky130_fd_pr__nfet_01v8 l={l} w={inv_wn} as={inv_wn*2*l} ad={i>
.ends INVX1

* NOR ckt-
.subckt NOR2 A B VDD VSS Z
xm01 NINT A VDD VDD sky130_fd_pr__pfet_01v8 l={l} w={nor_wp} as={nor_wp*2*l} ad>
xm02 Z B NINT VDD sky130_fd_pr__pfet_01v8 l={l} w={nor_wp} as={nor_wp*2*l} ad={>
xm03 Z A VSS VSS sky130_fd_pr__nfet_01v8 l={l} w={nor_wn} as={nor_wn*2*l} ad={n>
xm04 Z B VSS VSS sky130_fd_pr__nfet_01v8 l={l} w={nor_wn} as={nor_wn*2*l} ad={n>
.ends NOR2


VDD vdd 0 DC 1.8
VinA A 0 PULSE(0 1.8 0 20p 20p 1n 2n)
VinB B 0 DC 0


XNOR A B vdd 0 Z NOR2
XLOAD Z vdd 0 out2 INVX1

.tran 1p 4n

.control
run

plot V(A) V(B) V(Z)

meas tran trise trig V(Z) val=0.36 rise=1 targ V(Z) val=1.44 rise=1
meas tran tfall trig V(Z) val=1.44 fall=1 targ V(Z) val=0.36 fall=1
meas tran tpLH trig V(A) val=0.9 fall=1 targ V(Z) val=0.9 rise=1
meas tran tpHL trig V(A) val=0.9 rise=1 targ V(Z) val=0.9 fall=1

let tp=(tpLH+tpHL)/2
print trise tfall tpLH tpHL tp

.endc
.end
```

Table 11: 2-input NOR Rise and Fall Times from Schematic Simulation

Parameter	Value
Rise time t_r	21.74 ps
Fall time t_f	21.47 ps

4.2 Circuit Diagram

Provided in Figure 7.

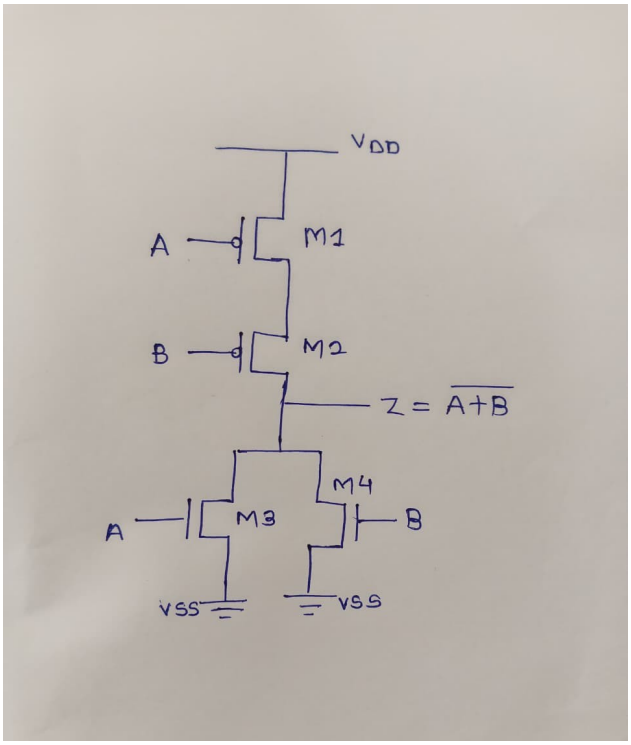


Figure 5: Circuit of NOR

Table 12: MOSFET Widths for the NOR Gate

MOSFET	Width (μm)
M1 and M2 (PMOS)	6 um
M3 and M4 (NMOS)	1.5

4.3 Magic Layout (DRC Clean)

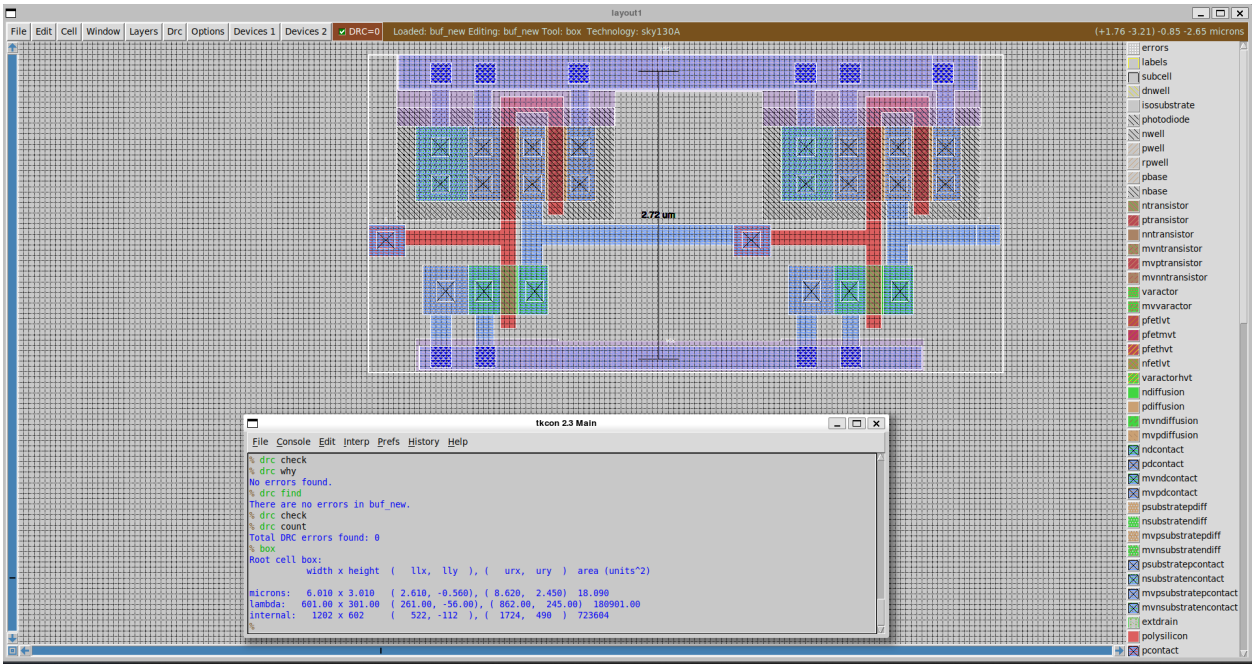


Figure 6: Magic Layout of Buffer Cell with width and height

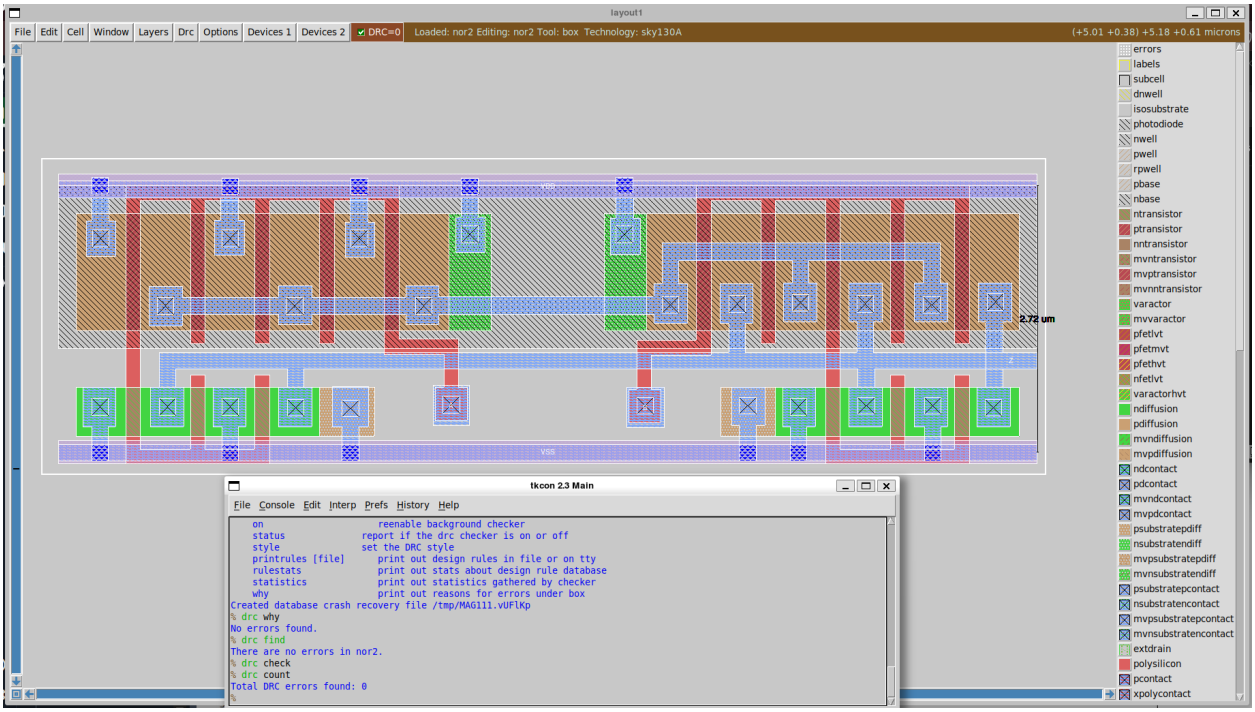


Figure 7: Magic Layout of NOR2 Cell with DRC Check

4.4 LVS check

```
/foss/designs > netgen -batch lvs "nor2_layout_fixed.spice nor2" "nor_netlist_schematic.spice NOR2" \
  /foss/pdks/sky130A/libs.tech/netgen/sky130A_setup.tcl lvs_report.out
Netgen 1.5.323 compiled on Tue Aug 25 11:33:45 AM CEST 2026
Warning: netgen command 'format' use fully-qualified name '::netgen::format'
Warning: netgen command 'global' use fully-qualified name '::netgen::global'
Reading netlist file nor2_layout_fixed.spice for nor2_layout_fixed.spice nor2
Call to undefined subcircuit sky130_fd_pr__pfet_01v8
Creating placeholder cell definition.
Call to undefined subcircuit sky130_fd_pr__nfet_01v8
Creating placeholder cell definition.
Reading netlist file nor_netlist_schematic.spice for nor_netlist_schematic.spice NOR2
Call to undefined subcircuit sky130_fd_pr__pfet_01v8
Creating placeholder cell definition.
Call to undefined subcircuit sky130_fd_pr__nfet_01v8
Creating placeholder cell definition.

Reading setup file /foss/pdks/sky130A/libs.tech/netgen/sky130A_setup.tcl

Model sky130_fd_pr__nfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__nfet_01v8
No property sa found for device sky130_fd_pr__nfet_01v8
No property sb found for device sky130_fd_pr__nfet_01v8
No property sd found for device sky130_fd_pr__nfet_01v8
No property nf found for device sky130_fd_pr__nfet_01v8
No property nrd found for device sky130_fd_pr__nfet_01v8
No property nrs found for device sky130_fd_pr__nfet_01v8
No property area found for device sky130_fd_pr__nfet_01v8
No property perim found for device sky130_fd_pr__nfet_01v8
No property topography found for device sky130_fd_pr__nfet_01v8
Model sky130_fd_pr__nfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__nfet_01v8
No property sa found for device sky130_fd_pr__nfet_01v8
No property sb found for device sky130_fd_pr__nfet_01v8
No property sd found for device sky130_fd_pr__nfet_01v8
No property nf found for device sky130_fd_pr__nfet_01v8
No property nrd found for device sky130_fd_pr__nfet_01v8
No property nrs found for device sky130_fd_pr__nfet_01v8
No property area found for device sky130_fd_pr__nfet_01v8
No property perim found for device sky130_fd_pr__nfet_01v8
No property topography found for device sky130_fd_pr__nfet_01v8
Model sky130_fd_pr__pfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__pfet_01v8
No property sa found for device sky130_fd_pr__pfet_01v8
No property sb found for device sky130_fd_pr__pfet_01v8
No property sd found for device sky130_fd_pr__pfet_01v8
No property nf found for device sky130_fd_pr__pfet_01v8
No property nrd found for device sky130_fd_pr__pfet_01v8
No property nrs found for device sky130_fd_pr__pfet_01v8
No property area found for device sky130_fd_pr__pfet_01v8
No property perim found for device sky130_fd_pr__pfet_01v8
```

```
No property perim found for device sky130_fd_pr__pfet_01v8
No property topography found for device sky130_fd_pr__pfet_01v8
Model sky130_fd_pr__pfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__pfet_01v8
No property sa found for device sky130_fd_pr__pfet_01v8
No property sb found for device sky130_fd_pr__pfet_01v8
No property sd found for device sky130_fd_pr__pfet_01v8
No property nf found for device sky130_fd_pr__pfet_01v8
No property nrd found for device sky130_fd_pr__pfet_01v8
No property nrs found for device sky130_fd_pr__pfet_01v8
No property area found for device sky130_fd_pr__pfet_01v8
No property perim found for device sky130_fd_pr__pfet_01v8
No property topography found for device sky130_fd_pr__pfet_01v8
Comparison output logged to file lvs_report.out
Logging to file "lvs_report.out" enabled
Circuit sky130_fd_pr__pfet_01v8 contains no devices.
Circuit sky130_fd_pr__nfet_01v8 contains no devices.

Contents of circuit 1:  Circuit: 'nor2'
Circuit nor2 contains 16 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:  10
  Class: sky130_fd_pr__nfet_01v8 instances:   6
Circuit contains 6 nets.
Contents of circuit 2:  Circuit: 'NOR2'
Circuit NOR2 contains 4 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:   2
  Class: sky130_fd_pr__nfet_01v8 instances:   2
Circuit contains 6 nets.

Circuit was modified by parallel/series device merging.
New circuit summary:

Contents of circuit 1:  Circuit: 'nor2'
Circuit nor2 contains 4 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:   2
  Class: sky130_fd_pr__nfet_01v8 instances:   2
Circuit contains 6 nets.
Contents of circuit 2:  Circuit: 'NOR2'
Circuit NOR2 contains 4 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:   2
  Class: sky130_fd_pr__nfet_01v8 instances:   2
Circuit contains 6 nets.

Circuit 1 contains 4 devices, Circuit 2 contains 4 devices.
Circuit 1 contains 6 nets,    Circuit 2 contains 6 nets.


Final result:
Circuits match uniquely.
```

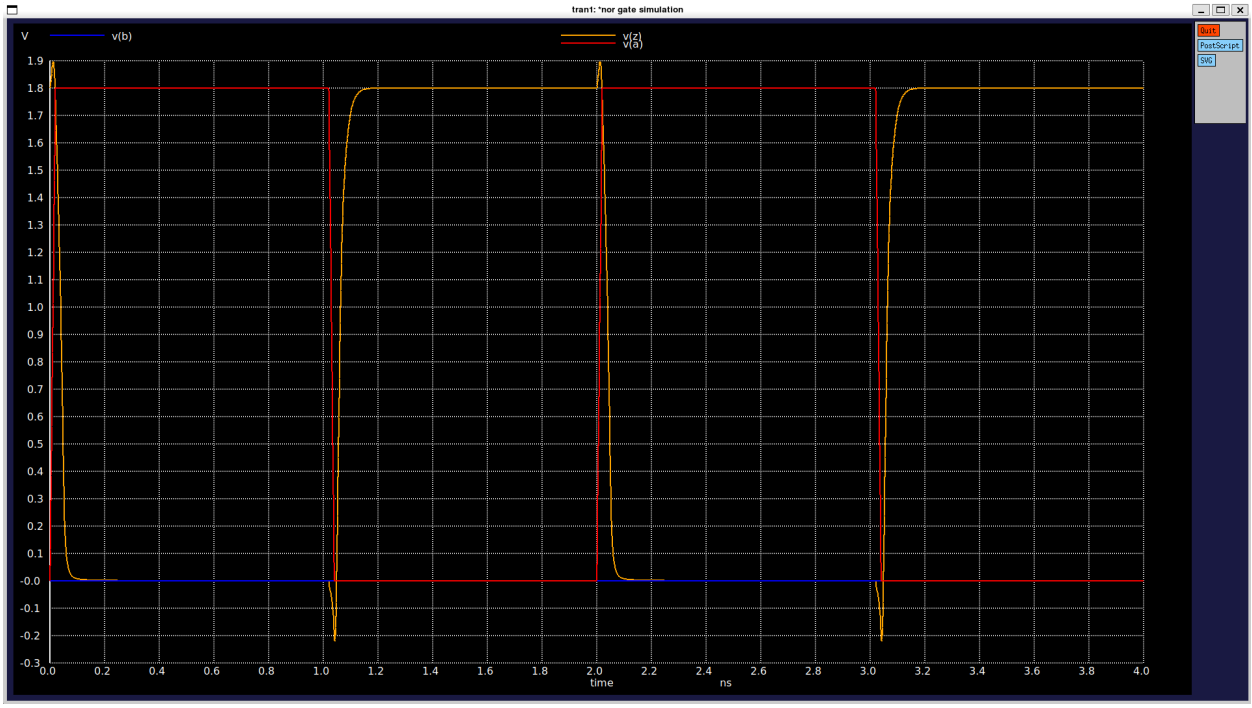
```
.
Logging to file "lvs_report.out" disabled
LVS Done.
```

4.5 PEX

```
* NGSPICE file created from nor2 (1).ext - technology: sky130A

X0 VDD A a_168_248# VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X1 Z B a_168_248# VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X2 VSS A Z VSS sky130_fd_pr__nfet_01v8 ad=0.17 pd=1.34667 as=0.17 ps=1.34667 w=0.5 l=0.15
X3 Z B VSS VSS sky130_fd_pr__nfet_01v8 ad=0.17 pd=1.34667 as=0.17 ps=1.34667 w=0.5 l=0.15
X4 VDD A a_168_248# VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X5 Z B VSS VSS sky130_fd_pr__nfet_01v8 ad=0.17 pd=1.34667 as=0.17 ps=1.34667 w=0.5 l=0.15
X6 Z A VSS VSS sky130_fd_pr__nfet_01v8 ad=0.17 pd=1.34667 as=0.17 ps=1.34667 w=0.5 l=0.15
X7 a_168_248# A VDD VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X8 Z A VSS VSS sky130_fd_pr__nfet_01v8 ad=0.17 pd=1.34667 as=0.17 ps=1.34667 w=0.5 l=0.15
X9 Z B a_168_248# VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X10 a_168_248# B Z VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X11 a_168_248# A VDD VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X12 a_168_248# B Z VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X13 Z B a_168_248# VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
X14 VSS B Z VSS sky130_fd_pr__nfet_01v8 ad=0.17 pd=1.34667 as=0.17 ps=1.34667 w=0.5 l=0.15
X15 a_168_248# A VDD VDD sky130_fd_pr__pfet_01v8 ad=0.3672 pd=2.052 as=0.3672 ps=2.052 w=1.2 l=0.15
C0 B VDD 0.40073f
C1 B Z 0.14547f
C2 a_168_248# A 0.06883f
C3 A B 0.00809f
C4 a_168_248# B 0.09284f
C5 Z VDD 0.13281f
C6 A VDD 0.45459f
C7 a_168_248# VDD 0.45268f
C8 A Z 0.08524f
C9 a_168_248# Z 0.3266f
C10 Z VSS 0.63512f
C11 a_168_248# VSS 0.06076f
C12 B VSS 0.75976f
C13 A VSS 0.76004f
C14 VDD VSS 2.79388f
/foss/designs >
```

4.6 Waveforms



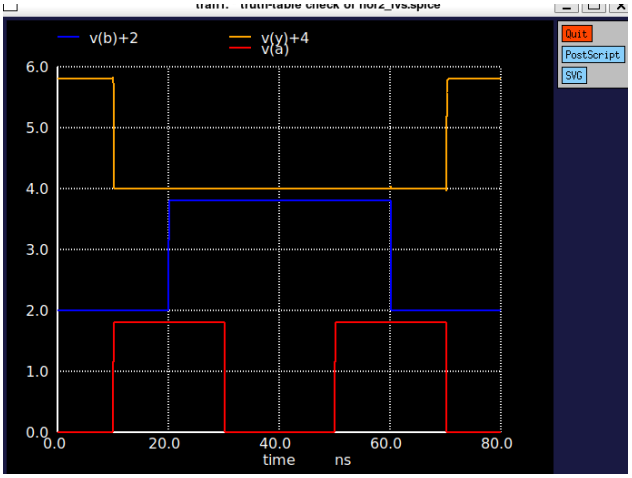
4.7 Testbench Simulation

```
/foss/designs > cat tb_nor2_layout.spice
* Truth-table check of nor2_lvs.spice
.lib $PDK_ROOT/sky130A/libs.tech/combined/sky130.lib.spice tt
.include nor2_layout_fixed.spice

Vdd VDD 0 1.8
XDUT A B VDD 0 Y nor2

Va A 0 pulse(0 1.8 10n 0.1n 0.1n 20n 40n)
Vb B 0 pulse(0 1.8 20n 0.1n 0.1n 40n 80n)
Cl Y 0 10f

.tran 0.1n 80n
.control
run
plot v(A) v(B)+2 v(Y)+4
.endc
.end
```



4.8 Capacitance considerations

Table 13: Input Pin Capacitance for the NOR gate

Input Pin	Rise Cap (pF)	Fall Cap (pF)	Average Cap (pF)
A (B tied to 0)	0.01057	0.01064	0.01061
B (A tied to 0)	0.01076	0.01076	0.01076

4.9 Timing considerations

4.9.1 Pin A (we fix B to logic-low)

Table 14: Output rise time (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.02480	0.02713	0.10002
10 fF	0.05044	0.05100	0.12787
100 fF	0.29412	0.29412	0.33597

Table 15: Output fall time (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.02618	0.02707	0.09486
10 fF	0.04483	0.04479	0.12794
100 fF	0.21641	0.21641	0.28518

Table 16: Cell rise propagation delay (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.03443	0.05365	0.09754
10 fF	0.05312	0.07277	0.15165
100 fF	0.22429	0.24465	0.40536

Table 17: Cell fall propagation delay (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.03507	0.05346	0.14386
10 fF	0.05063	0.06911	0.18312
100 fF	0.18330	0.20248	0.40024

4.9.2 Pin B (we fix A to logic-low)

Table 18: Output rise time (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.02393	0.03210	0.10021
10 fF	0.05023	0.05239	0.14566
100 fF	0.29412	0.29412	0.36602

Table 19: Output fall time (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.01327	0.01940	0.08258
10 fF	0.03132	0.03341	0.11812
100 fF	0.20250	0.20250	0.27944

Table 20: Cell rise propagation delay (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.02092	0.03592	0.07007
10 fF	0.03991	0.05639	0.12824
100 fF	0.21137	0.22852	0.40985

Table 21: Cell fall propagation delay (ns) vs. Input Slew and Load Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	0.02074	0.03679	0.08045
10 fF	0.03552	0.05398	0.13351
100 fF	0.16587	0.18650	0.37768

4.10 Power considerations

Table 22: Input combination vs. Static power

AB	Static power (nW)
00	0.0202
01	3.0614
10	1.3221
11	0.0584

4.10.1 Pin A (we fix B to logic-low)

Table 23: Dynamic rise power (nW) vs. Input Slew and Output Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	5140.64	5023.71	5223.37
10 fF	8249.80	8141.47	8237.86
100 fF	37431.00	37395.18	37342.98

Table 24: Dynamic fall power (nW) vs. Input Slew and Output Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	1061.62	766.81	983.66
10 fF	1063.89	809.55	914.68
100 fF	1065.39	901.55	808.95

4.10.2 Pin B (we fix A to logic-low)

Table 25: Dynamic rise power (nW) vs. Input Slew and Output Capacitance

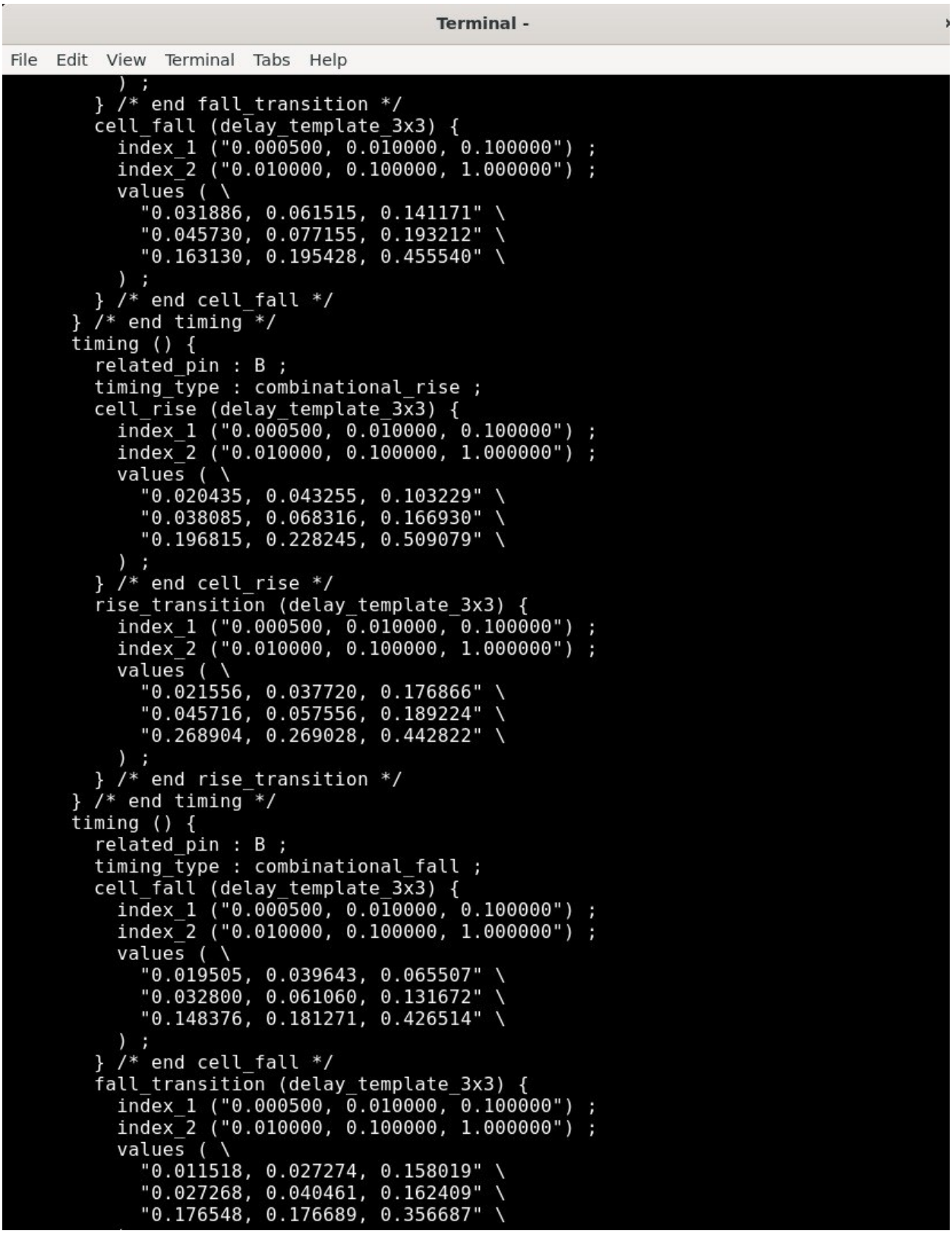
	10 ps	100 ps	1000 ps
0.5 fF	3829.82	3687.57	4814.41
10 fF	6976.96	6817.03	7680.35
100 fF	36177.30	36124.92	36369.72

Table 26: Dynamic fall power (nW) vs. Input Slew and Output Capacitance

	10 ps	100 ps	1000 ps
0.5 fF	593.68	917.24	241.00
10 fF	477.84	752.70	520.39
100 fF	396.80	385.99	659.33

4.11 CharLib

```
Terminal -
File  Edit  View  Terminal  Tabs  Help
/foss/designs/charlib > grep -n "cell_rise\|cell_fall\|rise_transition\|fall_transi
tion" results/sky130_nor2.lib
49:        cell_rise (delay_template_3x3) {
57:        } /* end cell_rise */
58:        rise_transition (delay_template_3x3) {
66:        } /* end rise_transition */
71:        fall_transition (delay_template_3x3) {
79:        } /* end fall_transition */
80:        cell_fall (delay_template_3x3) {
88:        } /* end cell_fall */
93:        cell_rise (delay_template_3x3) {
101:        } /* end cell_rise */
102:        rise_transition (delay_template_3x3) {
110:        } /* end rise_transition */
115:        cell_fall (delay_template_3x3) {
123:        } /* end cell_fall */
124:        fall_transition (delay_template_3x3) {
132:        } /* end fall_transition */
/foss/designs/charlib > sed -n '45,135p' results/sky130_nor2.lib
      function : "Z = !(A|B)" ;
      timing () {
        related_pin : A ;
        timing_type : combinational_rise ;
        cell_rise (delay_template_3x3) {
          index_1 ("0.000500, 0.010000, 0.100000") ;
          index_2 ("0.010000, 0.100000, 1.000000") ;
          values ( \
            "0.033623, 0.063860, 0.124580" \
            "0.050971, 0.083634, 0.190486" \
            "0.209309, 0.242914, 0.496639" \
          ) ;
        } /* end cell_rise */
        rise_transition (delay_template_3x3) {
          index_1 ("0.000500, 0.010000, 0.100000") ;
          index_2 ("0.010000, 0.100000, 1.000000") ;
          values ( \
            "0.022535, 0.031199, 0.175642" \
            "0.045935, 0.050708, 0.171852" \
            "0.268904, 0.269038, 0.386913" \
          ) ;
        } /* end rise_transition */
      } /* end timing */
      timing () {
        related_pin : A ;
        timing_type : combinational_fall ;
        fall_transition (delay_template_3x3) {
          index_1 ("0.000500, 0.010000, 0.100000") ;
          index_2 ("0.010000, 0.100000, 1.000000") ;
          values ( \
            "0.022219, 0.033286, 0.157728" \
            "0.038663, 0.045453, 0.180177" \
            "0.188581, 0.188745, 0.363135" \
          ) ;
```


Terminal -
File Edit View Terminal Tabs Help
) ;
} /* end fall_transition */
cell_fall (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.031886, 0.061515, 0.141171" \
"0.045730, 0.077155, 0.193212" \
"0.163130, 0.195428, 0.455540" \
) ;
} /* end cell_fall */
} /* end timing */
timing () {
related_pin : B ;
timing_type : combinational_rise ;
cell_rise (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.020435, 0.043255, 0.103229" \
"0.038085, 0.068316, 0.166930" \
"0.196815, 0.228245, 0.509079" \
) ;
} /* end cell_rise */
rise_transition (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.021556, 0.037720, 0.176866" \
"0.045716, 0.057556, 0.189224" \
"0.268904, 0.269028, 0.442822" \
) ;
} /* end rise_transition */
} /* end timing */
timing () {
related_pin : B ;
timing_type : combinational_fall ;
cell_fall (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.019505, 0.039643, 0.065507" \
"0.032800, 0.061060, 0.131672" \
"0.148376, 0.181271, 0.426514" \
) ;
} /* end cell_fall */
fall_transition (delay_template_3x3) {
index_1 ("0.000500, 0.010000, 0.100000") ;
index_2 ("0.010000, 0.100000, 1.000000") ;
values (\
"0.011518, 0.027274, 0.158019" \
"0.027268, 0.040461, 0.162409" \
"0.176548, 0.176689, 0.356687" \

5 Sequential: Positive edge-triggered D-flipflop with single output

5.1 Schematic Design

Table 27: DFF internal gate Rise and Fall Times from Schematic Simulation

Parameter	Value
Rise time t_r	19.83 ps
Fall time t_f	19.91 ps

5.2 Circuit Diagram

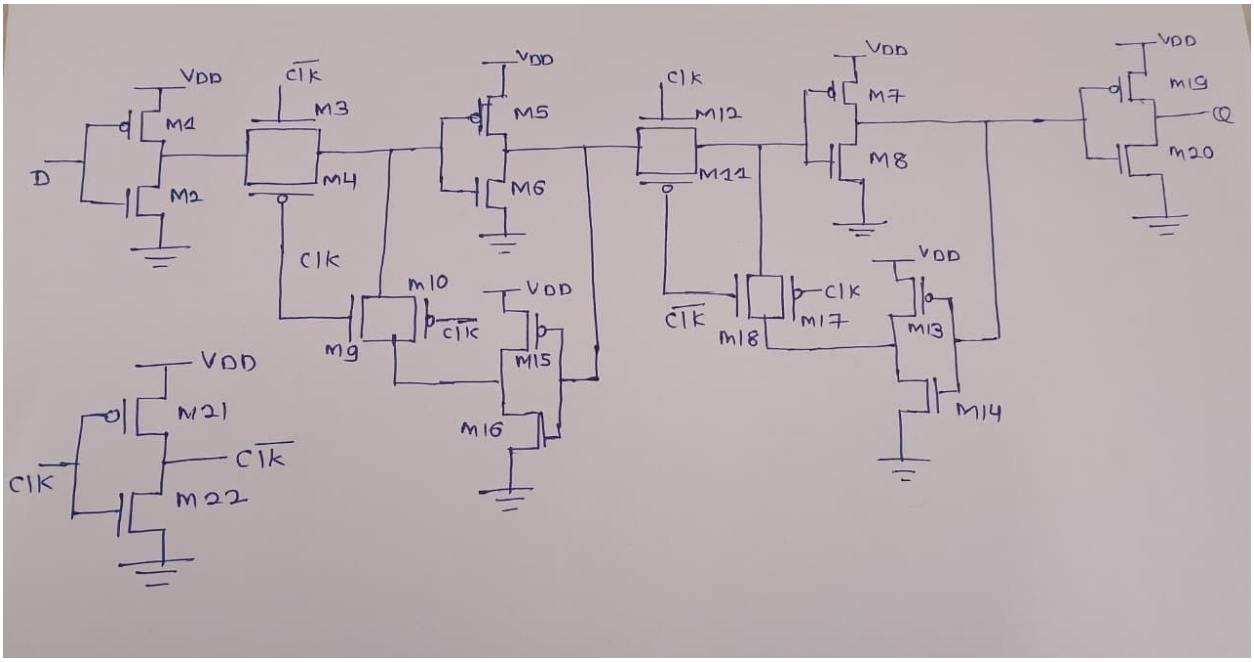


Table 28: MOSFET Widths for the DFF

MOSFET	Width (μm)
M1, M5, M7, M21	2.4
M2, M6, M8, M22	0.8
M15, M13	1.24
M16, M14	0.42
M19	3.4
M20	1.0
M4, M10, M11, M17	1.24
M3, M9, M12, M18	0.42

```
.lib /foss/pdks/sky130A/libs.tech/ngspice/sky130.lib.spice tt

```

```
3
4  * INVX1
5  * pins: input VDD VSS output
6
7
8  .subckt INVX1 A VDD VSS Y
9  xm01 Y A VDD VDD sky130_fd_pr__pfet_01v8 l=0.15 w=2.4 as=0.72 ad=0.72 ps=5.4 pd=5.4
10 xm02 Y A VSS VSS sky130_fd_pr__nfet_01v8 l=0.15 w=0.8 as=0.24 ad=0.24 ps=2.2 pd=2.2
11 .ends
12
13 .subckt MIN_INV A VDD VSS Y
14 xm01 Y A VDD VDD sky130_fd_pr__pfet_01v8 l=0.15 w=1.24 as=0.372 ad=0.372 ps=3.08 pd=3.08
15 xm02 Y A VSS VSS sky130_fd_pr__nfet_01v8 l=0.15 w=0.42 as=0.126 ad=0.126 ps=1.44 pd=1.44
16 .ends
17
18
19 * TRANSMISSION GATE
20 *
21 * pins:
22 * IN OUT VDD VSS CTRL CTRL_B
23 *
24 * NMOS controlled by CTRL
25 * PMOS controlled by CTRL_B
26 *=========================================================
27
28 .subckt TG IN OUT VDD VSS CTRL CTRL_B
29 XM_N OUT CTRL IN VSS sky130_fd_pr__nfet_01v8 l=0.15 w=0.42 as=0.402 ad=0.402 ps=3.28 pd=3.28
30 XM_P OUT CTRL_B IN VDD sky130_fd_pr__pfet_01v8 l=0.15 w=1.24 as=0.372 ad=0.372 ps=3.08 pd=3.08
31 .ends
32
33 .subckt DFF_OUTINV A VDD VSS Y
34
35 XM_P Y A VDD VDD sky130_fd_pr__pfet_01v8 l=0.15 w=3.4 as=1.02 ad=1.02 ps=7.4 pd=7.4
36 XM_N Y A VSS VSS sky130_fd_pr__nfet_01v8 l=0.15 w=1 as=0.3 ad=0.3 ps=2.6 pd=2.6
37
38 .ends DFF_OUTINV
39
40
41 * DFXTP CELL
42 * pins: D VDD VSS CLK Q
43
44
45 .subckt DFXTP D VDD VSS CLK Q
46 XCLKINV CLK VDD VSS CLKB INVX1
47
48 * MASTER LATCH
49 XI1 D VDD VSS D_bar INVX1
50 XTG1 D_bar M VDD VSS CLKB CLK TG
51 XI2 M VDD VSS Yint INVX1
52 XI3 Yint VDD VSS M_bar MIN_INV
53 XTG2 M_bar M VDD VSS CLK CLKB TG
54 XTG3 Yint S VDD VSS CLK CLKB TG
55 XI5 S VDD VSS X INVX1
56 XI6 X VDD VSS S_bar MIN_INV
57 XTG4 S_bar S VDD VSS CLKB CLK TG
58 XI4 X VDD VSS Q DFF_OUTINV
59 .ends DFXTP
60
61 * TESTBENCH
```

```
62
63 VDD vdd 0 DC 1.8
64 VD d 0 PULSE(0 1.8 0.5n 20p 20p 2n 4n)
65 VCLK clk 0 PULSE(0 1.8 0 20p 20p 1n 2n)
66
67 * DFXTP
68 XDFT d vdd 0 clk q DFXTP
69 Xload q vdd 0 load_out INVX1
70
71 .tran 1p 6n
72
73
74 .control
75 run
76 plot V(clk) V(d) V(q)
77 meas tran trise TRIG V(q) VAL=0.36 RISE=1 TARG V(q) VAL=1.44 RISE=1 FROM=1n
78 meas tran tfall TRIG V(q) VAL=1.44 FALL=1 TARG V(q) VAL=0.36 FALL=1 FROM=1n
79
80 print trise tfall
81 .endc
82 .end
```

5.3 Magic Layout (DRC Clean)

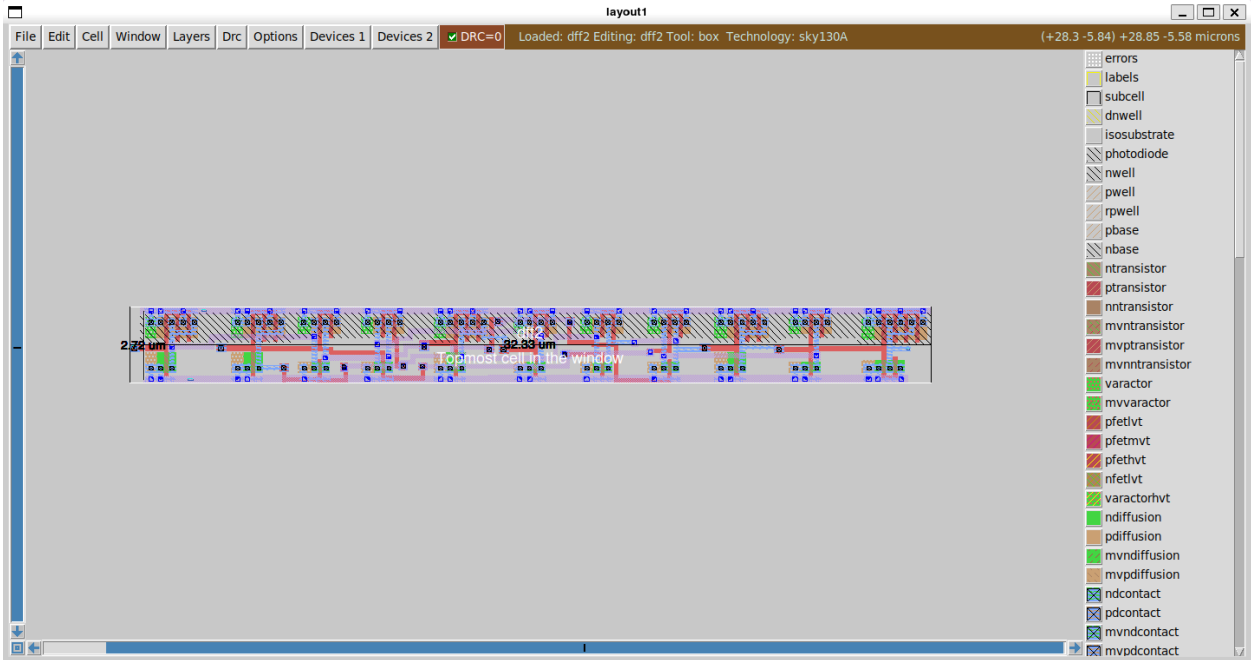


Figure 8: Magic Layout of DFF Cell with width and height

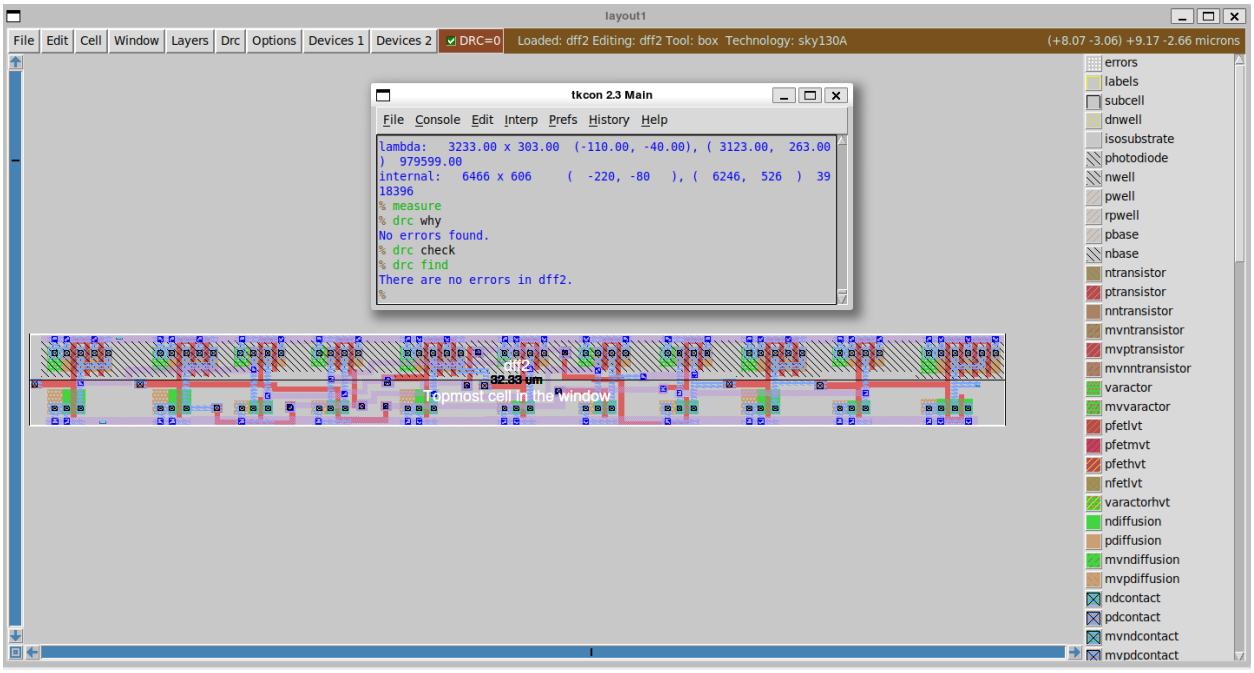


Figure 9: Magic Layout of DFF Cell with DRC Check

5.4 LVS check

```
No property topology found for device sky130_fd_pr__pfet_01v8
Comparison output logged to file comp.out
Logging to file "comp.out" enabled
Circuit sky130_fd_pr__pfet_01v8 contains no devices.
Circuit sky130_fd_pr__pfet_01v8 contains no devices.

Contents of circuit 1: Circuit: 'DFXTP'
Circuit dff_final_1 contains 28 device instances.
    Class: sky130_fd_pr__pfet_01v8 instances: 11
    Class: sky130_fd_pr__nfet_01v8 instances: 17
Circuit contains 13 nets.

Contents of circuit 2: Circuit: 'pdff'
Circuit dff_final_1 contains 22 device instances.
    Class: sky130_fd_pr__nfet_01v8 instances: 11
    Class: sky130_fd_pr__pfet_01v8 instances: 11
Circuit contains 13 nets.

Circuit was modified by parallel/series device merging.
New circuit summary:

Contents of circuit 1: Circuit: 'DFXTP'
Circuit dff_final_1 contains 22 device instances.
    Class: sky130_fd_pr__nfet_01v8 instances: 11
    Class: sky130_fd_pr__pfet_01v8 instances: 11
Circuit contains 13 nets.

Contents of circuit 2: Circuit: 'pdff'
Circuit dff_final_1 contains 22 device instances.
    Class: sky130_fd_pr__nfet_01v8 instances: 11
    Class: sky130_fd_pr__pfet_01v8 instances: 11
Circuit contains 13 nets.

Circuit 1 contains 22 devices, Circuit 2 contains 22 devices.
Circuit 1 contains 13 nets, Circuit 2 contains 13 nets.


Final result:
Circuits match uniquely.
Property errors were found.

The following cells had property errors:
dff_final_1

Logging to file "comp.out" disabled
LVS Done.
```

Figure 10: LVS done

5.5 PEX

```
* NGSPICE file created from dff_final_1.ext - technology: sky130A


X0 a_1356_102# a_886_102# a_344_102# GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1
    .44 w=0.42 l=0.15
X1 a_3556_102# a_886_102# a_2385_102# VDD sky130_fd_pr__pfet_01v8 ad=0.294 pd=2.56 as=0.19534
    ps=1.69691 w=0.98 l=0.15
X2 VDD a_4585_102# a_4098_110# VDD sky130_fd_pr__pfet_01v8 ad=0.3 pd=2.60139 as=0.19933 ps=1
    .73154 w=1 l=0.15
X3 a_4098_110# a_4585_102# GND GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w
    =0.42 l=0.15
X4 a_4098_110# a_4585_102# VDD VDD sky130_fd_pr__pfet_01v8 ad=0.19933 pd=1.73154 as=0.3 ps=2
    .60139 w=1 l=0.15
X5 a_1356_102# CLK a_344_102# VDD sky130_fd_pr__pfet_01v8 ad=0.294 pd=2.56 as=0.19534 ps=1
    .69691 w=0.98 l=0.15
X6 Q a_4585_102# GND GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0
    .15
X7 a_4098_110# a_886_102# a_3556_102# GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps
    =1.44 w=0.42 l=0.15
X8 a_4098_110# CLK a_3556_102# VDD sky130_fd_pr__pfet_01v8 ad=0.19534 pd=1.69691 as=0.294 ps=2
    .56 w=0.98 l=0.15
X9 a_1898_110# a_2385_102# GND GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w
    =0.42 l=0.15
```

```
X10 VDD a_2385_102# a_1898_110# VDD sky130_fd_pr__pfet_01v8 ad=0.3 pd=2.60139 as=0.19933 ps=1
    .73154 w=1 l=0.15
X11 a_886_102# CLK GND GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l
    =0.15
X12 VDD CLK a_886_102# VDD sky130_fd_pr__pfet_01v8 ad=0.3 pd=2.60139 as=0.15 ps=1.3 w=1 l=0.15
X13 a_1898_110# a_2385_102# VDD VDD sky130_fd_pr__pfet_01v8 ad=0.19933 pd=1.73154 as=0.3 ps=2
    .60139 w=1 l=0.15
X14 a_886_102# CLK VDD VDD sky130_fd_pr__pfet_01v8 ad=0.15 pd=1.3 as=0.3 ps=2.60139 w=1 l=0.15
X15 VDD a_3556_102# a_4585_102# VDD sky130_fd_pr__pfet_01v8 ad=0.3 pd=2.60139 as=0.15 ps=1.3 w
    =1 l=0.15
X16 a_4585_102# a_3556_102# GND GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w
    =0.42 l=0.15
X17 a_344_102# D GND GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0
    .15
X18 a_4585_102# a_3556_102# VDD VDD sky130_fd_pr__pfet_01v8 ad=0.15 pd=1.3 as=0.3 ps=2.60139 w
    =1 l=0.15
X19 a_2385_102# a_1356_102# GND GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w
    =0.42 l=0.15
X20 VDD D a_344_102# VDD sky130_fd_pr__pfet_01v8 ad=0.3 pd=2.60139 as=0.19933 ps=1.73154 w=1 l
    =0.15
X21 VDD a_1356_102# a_2385_102# VDD sky130_fd_pr__pfet_01v8 ad=0.3 pd=2.60139 as=0.19933 ps=1
    .73154 w=1 l=0.15
X22 Q a_4585_102# VDD VDD sky130_fd_pr__pfet_01v8 ad=0.291 pd=2.54 as=0.291 ps=2.52335 w=0.97
    l=0.15
X23 a_344_102# D VDD VDD sky130_fd_pr__pfet_01v8 ad=0.19933 pd=1.73154 as=0.3 ps=2.60139 w=1 l
    =0.15
X24 a_2385_102# a_1356_102# VDD VDD sky130_fd_pr__pfet_01v8 ad=0.19933 pd=1.73154 as=0.3 ps=2
    .60139 w=1 l=0.15
X25 a_1898_110# CLK a_1356_102# GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w
    =0.42 l=0.15
X26 a_3556_102# CLK a_2385_102# GND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w
    =0.42 l=0.15
X27 a_1898_110# a_886_102# a_1356_102# VDD sky130_fd_pr__pfet_01v8 ad=0.19534 pd=1.69691 as=0
    .294 ps=2.56 w=0.98 l=0.15
C0 a_3556_102# a_2385_102# 0.11762f
C1 a_4585_102# a_886_102# 0.05317f
C2 a_4585_102# Q 0.07174f
C3 a_4585_102# CLK 0.04649f
C4 a_4098_110# VDD 0.57688f
C5 a_2385_102# a_886_102# 0.07665f
C6 a_2385_102# CLK 0.08447f
C7 a_3556_102# VDD 0.39022f
C8 a_1898_110# a_1356_102# 0.29427f
C9 a_886_102# VDD 0.46108f
C10 Q VDD 0.07673f
C11 CLK VDD 0.37709f
C12 D VDD 0.02808f
C13 a_344_102# a_886_102# 0.23081f
C14 a_344_102# CLK 0.15518f
C15 a_344_102# D 0.07652f
C16 a_4585_102# VDD 0.20959f
C17 a_2385_102# VDD 0.3036f
C18 a_3556_102# a_1898_110# 0
C19 a_886_102# a_1356_102# 0.13315f
C20 a_1356_102# CLK 0.24603f
C21 a_1898_110# a_886_102# 0.04283f
```

```
C22 a_1898_110# CLK 0.01963f
C23 a_3556_102# a_4098_110# 0.29427f
C24 a_344_102# VDD 0.4849f
C25 a_2385_102# a_1356_102# 0.08837f
C26 a_886_102# a_4098_110# 0.01875f
C27 a_4098_110# Q 0.01594f
C28 a_4098_110# CLK 0.02966f
C29 a_1898_110# a_2385_102# 0.46895f
C30 a_3556_102# a_886_102# 0.24603f
C31 a_3556_102# CLK 0.1324f
C32 a_1356_102# VDD 0.39038f
C33 a_886_102# Q 0
C34 a_886_102# CLK 0.51111f
C35 a_4585_102# a_4098_110# 0.47735f
C36 Q CLK 0
C37 a_886_102# D 0
C38 D CLK 0.01186f
C39 a_1898_110# VDD 0.59118f
C40 a_4585_102# a_3556_102# 0.08837f
C41 a_344_102# a_1356_102# 0.11964f
C42 Q GND 0.1831f
C43 D GND 0.43815f
C44 CLK GND 3.09061f
C45 VDD GND 6.17278f
C46 a_4585_102# GND 1.01306f
C47 a_4098_110# GND 0.27452f
C48 a_3556_102# GND 0.6209f
C49 a_2385_102# GND 0.97287f
C50 a_1898_110# GND 0.25825f
C51 a_1356_102# GND 0.62093f
C52 a_344_102# GND 0.39141f
C53 a_886_102# GND 2.98792f
```

5.6 Waveforms

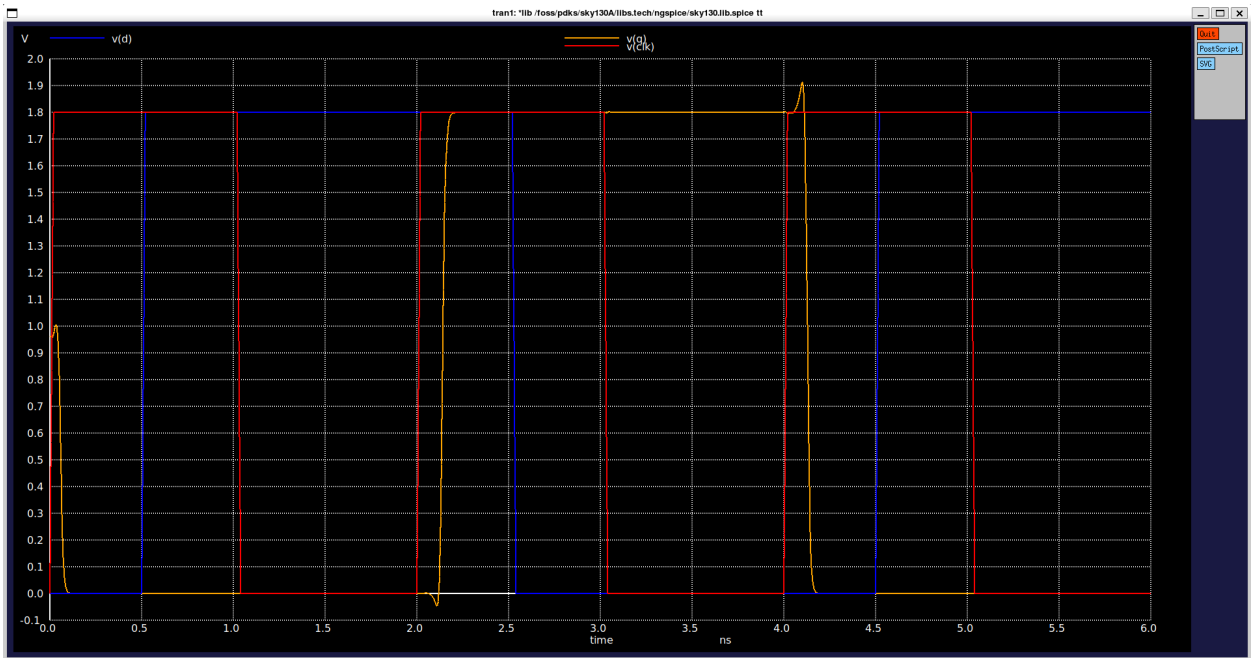


Figure 11: DFF input, output and clock waveforms showing rise and fall swings and propagation delays

6 Challenges Faced

One of the main challenges was achieving symmetric rise time (t_r) and fall time (t_f) in the inverter using NGSpice simulations. The PMOS and NMOS widths were adjusted to balance their driving strengths and obtain nearly equal t_r and t_f. After achieving the desired timing characteristics, these transistor dimensions were used for the layout design. Another major challenge was implementing the **finger approach** for the NMOS and PMOS transistors while satisfying the **2.72 μm height constraint**. The transistors had to be divided into multiple fingers and arranged carefully within the available height while maintaining proper source, drain, and gate connections. This required several iterations to achieve the desired layout while avoiding DRC errors and maintaining correct connectivity.

7 Contributions of Team Members

7.1 Aakanksha Tambare

1. Designing and layouts for DFF and Buffer
2. Setup and Hold Simulations for the D Flip Flop

7.2 Bhakti Shah

1. Designing Schematic, Timing and Power Characteri-zation of the NOR and Buffer gate
2. Verilog files for all the 3 cells

7.3 Rashi Mullick

1. Schematic, Layout and LVS of NOR2
2. CharLib Files

7.4 Sayali Mahajan

1. Schematic, Layout and LVS of NOR2
2. Rise and Fall time of NOR2